



Lakshya JEE 2.0 (2024)

Limit, Continuity and Differentiability

DPP-07

1. The function $y = |\sin x|$ is continuous for any x but it is not differentiable at
- (1) $x = 0$ only
 - (2) $x = \pi$ only
 - (3) $x = k\pi$ (k is an positive integer) only
 - (4) $x = 0$ and $x = k\pi$ (k is an integer) only
2. Let $f(x) = \begin{cases} 0, & x < 0 \\ x^2, & x \geq 0 \end{cases}$, then for all values for x
- (1) f is continuous but not differentiable
 - (2) f is differentiable but not continuous
 - (3) f' is continuous but not differentiable
 - (4) f' is continuous and differentiable
3. Let $f(x) = [x]$, then $f(x)$ is:
- (1) Differentiable for all $x \in (R - I)$
 - (2) Continuous for all $x \in R$
 - (3) Continuous no where
 - (4) Differentiable for all $x \in R$
4. If $f: R \rightarrow R$ is defined by $f(x) = \max\{x, x^3\}$, then the set of all points where $f(x)$ is not differentiable
- (1) $\{-1, 0, 1\}$
 - (2) $\{0, 1\}$
 - (3) ϕ
 - (4) $\{-1, 1\}$
5. If the function $g(x) = \begin{cases} k\sqrt{x+1}, & 0 \leq x \leq 3 \\ mx+2, & 3 < x \leq 5 \end{cases}$ is differentiable, then the value of $(k+m)$ is:
- (1) $\frac{10}{3}$
 - (2) 4
 - (3) 2
 - (4) $\frac{16}{5}$
6. Let $f(x) = \begin{cases} 2x^2 - x + 6, & x \leq 1 \\ -3x + 5x^2 + 5, & x > 1 \end{cases}$, then prove that $f(x)$ is non-differentiable at $x = 1$.
7. If $f(x) = e^{|x|}$ then which one of the following is correct?
- (1) $f'(0) = 1$
 - (2) $f'(0) = -1$
 - (3) $f'(0) = 0$
 - (4) $f'(0)$ does not exist
8. Consider the function $f(x) = \begin{cases} x^2, & x > 2 \\ 3x-2, & x \leq 2 \end{cases}$ Which one of the following statements is correct in respect of the above function?
- (1) $f(x)$ is derivable but not continuous at $x = 2$
 - (2) $f(x)$ is continuous but not derivable at $x = 2$
 - (3) $f(x)$ is neither continuous nor derivable at $x = 2$
 - (4) $f(x)$ is continuous as well as derivable at $x = 2$
9. If $f(x) = |x-3|$, then:
1. $f(x)$ is continuous at $x = 3$
 2. $f(x)$ is differentiable at $x = 0$
- Which of the above statements is/are correct?
- (1) 1 Only
 - (2) 2 Only
 - (3) Both 1 and 2
 - (4) Neither 1 nor 2
10. Comment on the differentiability of $f(x) = \begin{cases} 2x+3, & x < 1 \\ 4x^2-1, & x \geq 1 \end{cases}$ at $x = 1$.
11. If $f(x) = \begin{cases} [\sin \frac{3\pi}{2} x], & x \leq 1 \\ 2\{x\}-1, & x > 1 \end{cases}$, then comment on the derivability at $x = 1$, where $[.]$ is greatest integer function and $\{.\}$ is fractional part function.

12. If $f(x) = \begin{cases} \left[\frac{2x}{3} \right] + \frac{x}{3} + 2, & x < 3 \\ \left\{ \frac{x}{3} \right\} + 3, & x \geq 3 \end{cases},$

then comment on the continuity and differentiable at $x = 3$, where $[.]$ is greatest integer function and $\{.\}$ is fractional part function.

13. The function $f(x) = \max[(1 - x), (1 + x), 2]$, $x \in (-\infty, \infty)$ is
- (1) Continuous at all points
 - (2) Differentiable at all points
 - (3) Differentiable at all points except at $x = 1$ and $x = -1$
 - (4) None of the above

14. The number of points where

$$f(x) = \begin{cases} [\cos \pi x], & x \leq 1 \\ |2x - 3|[x - 2], & x > 1 \end{cases} \quad (\text{where } [.] \text{ denotes the greatest integer function})$$

in $[0, 2]$ is discontinuous is P and non-differentiable is Q then find $(P + Q)$

15. If $f(x) = \begin{cases} ax^2 - b, & |x| < 1 \\ \frac{1}{|x|}, & |x| \geq 1 \end{cases}$ is differentiable at

$x = 1$, then

- (1) $a = \frac{1}{2}, b = -\frac{1}{2}$
- (2) $a = -\frac{1}{2}, b = -\frac{3}{2}$
- (3) $a = b = \frac{1}{2}$
- (4) $a = b = -\frac{1}{2}$

Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

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| 1. (4) | 9. (3) |
| 2. (3) | 10. (not differentiable at $x=1$) |
| 3. (1) | 11. (not differentiable at $x=1$) |
| 4. (1) | 12. (Discontinuous and non-differentiable) |
| 5. (3) | 13. (1, 3) |
| 6. (Proof) | 14. (8) |
| 7. (4) | 15. (2) |
| 8. (2) | |



PW Web/App - <https://smart.link/7wwosivoicgd4>

Library- <https://smart.link/sdfez8ejd80if>